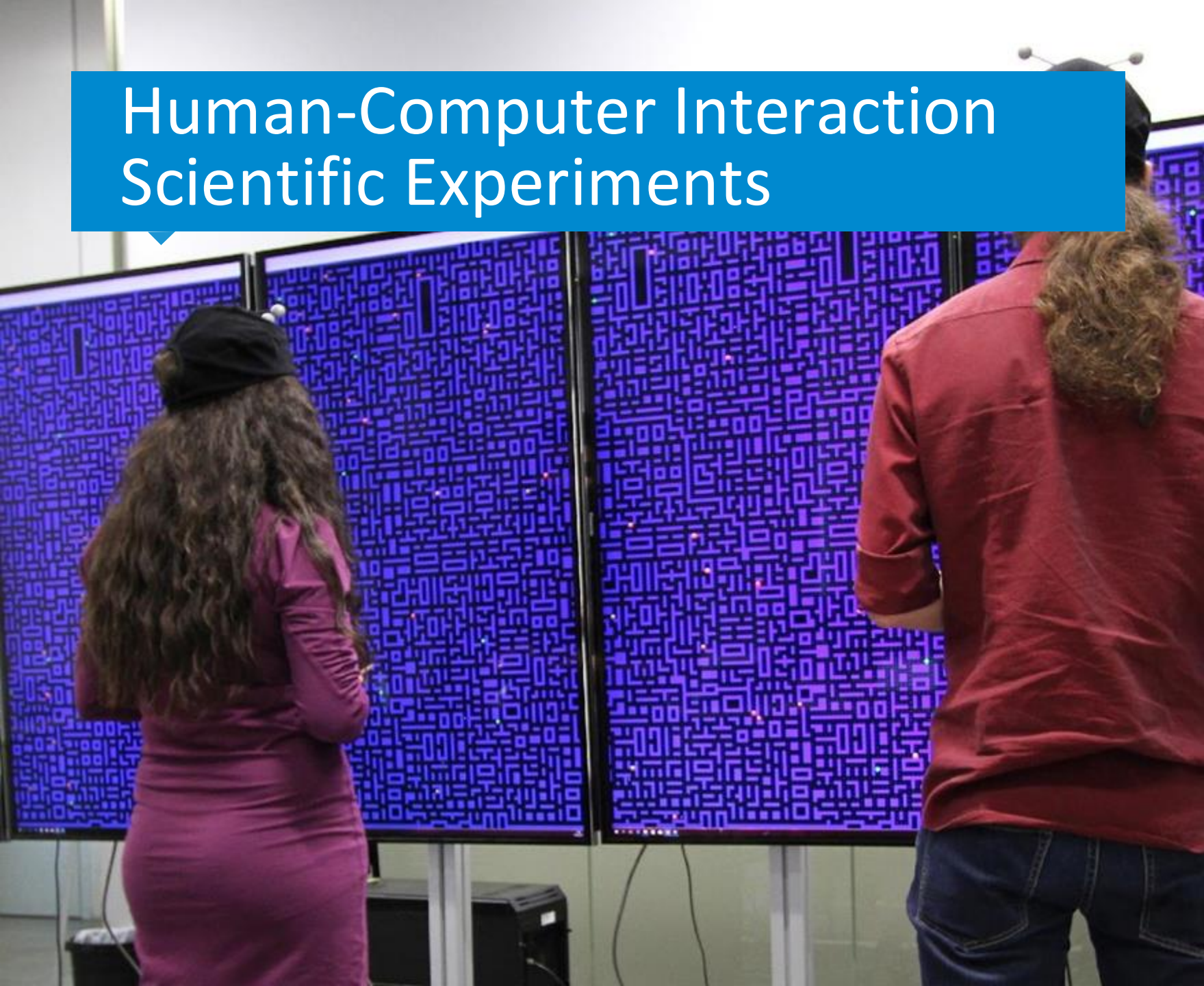


Human-Computer Interaction Scientific Experiments



Goals of this lecture

After this lecture you will be able to explain ...

- Why we conduct experiments
- What a hypothesis is
- How to operationalize a hypothesis
- Different research methods
- Independent and dependent variables
- Confounding variables
- Between-group design vs. within-group design

Goals of this lecture

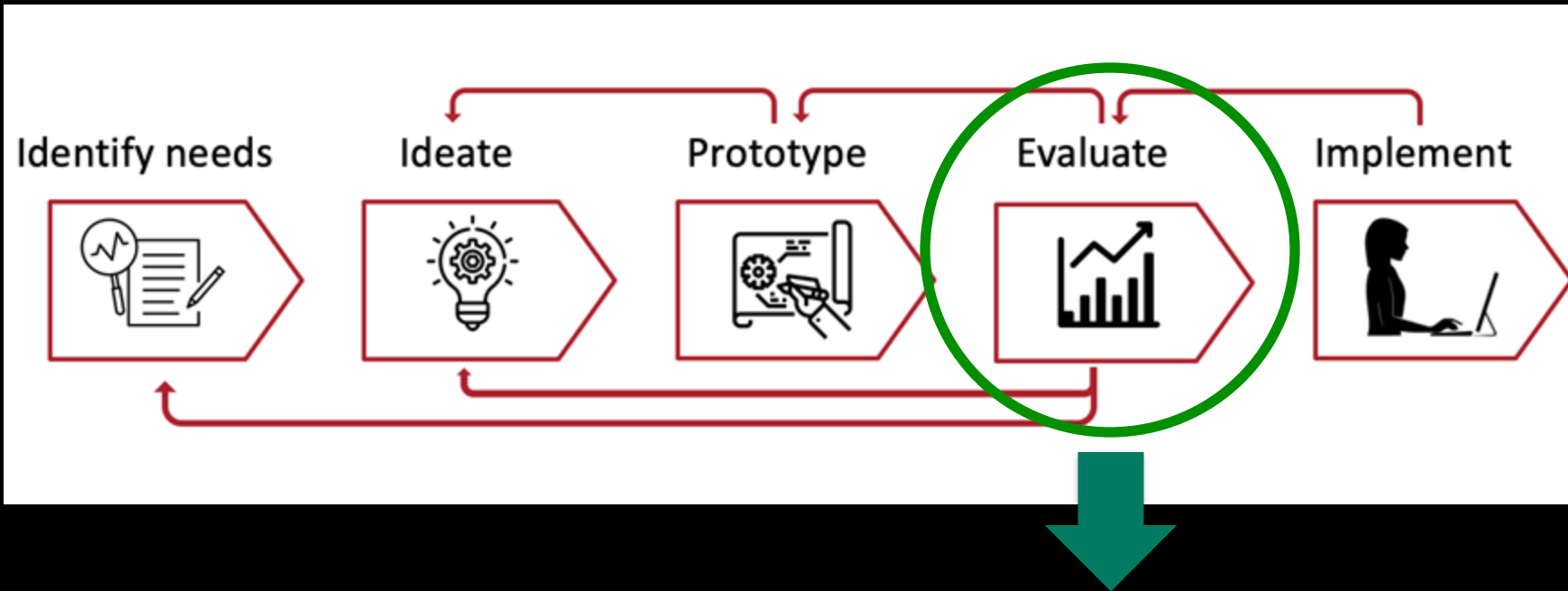
Chapter 4: “Scientific Foundations”:

- 4.1 → 4.8

Chapter 5: “Designing HCI Experiments”

- 5.1 → 5.14

HCI design process



We must conduct research

What is research?

Research is **investigation** or **experimentation** aimed at the **discovery** and **interpretation** of facts and **revision** of accepted theories or laws in light of new facts.



What is research?



Conceptual Research

Focuses on theoretical exploration and development of concepts and ideas

Deals with abstract and hypothetical concepts

Aims to generate new knowledge or refine existing theories

Does not involve direct observation or collection of empirical data

Provides a theoretical foundation for empirical research



Empirical Research

Focuses on gathering empirical evidence and testing hypotheses

Deals with real-world observations and data

Aims to test hypotheses, validate theories, or solve practical problems

Involves direct observation, data collection, and analysis of empirical data

Builds upon conceptual frameworks by testing and validating them

Why do we conduct research?

Scientists want to understand **cause** and **effect**

*When **metal** is heated it **expands***



Why do we conduct research?

To make **predictions**

*The metal in this bridge **needs space to expand** in hot weather*

Why do we conduct HCI research?

To test **usability and user experience**

Keyboard 1 allows faster input than keyboard 2

Users make less errors using interface X than Y

70% of older adults can use mobile app XX without instructions.

...

Why do we conduct HCI research?

Therefore, evaluation in HCI has three goals:

- 1) to identify any specific **problems or errors** in the design
- 2) to assess the **usability and accessibility** of the design
- 3) to assess **users' experience** with the design



Functionality



Usability



User experience (UX)

Are the two different?

Effectiveness
Efficiency
Learnability
Memorability
Error prevention



Usability

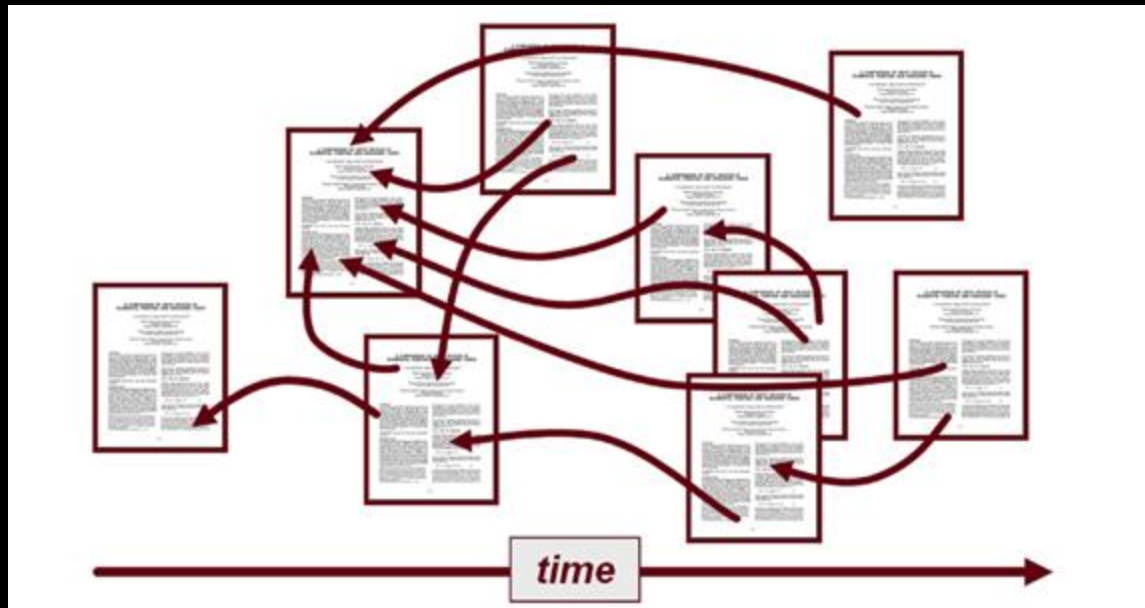
Satisfaction
Enjoyment
Pleasure
Fun
Value



User
experience

Research must be:

- *Reproducible (rigorous method)*
- *Peer reviewed and published (detailed report)*
- *Builds on previous scientific evidence (references & citations)*



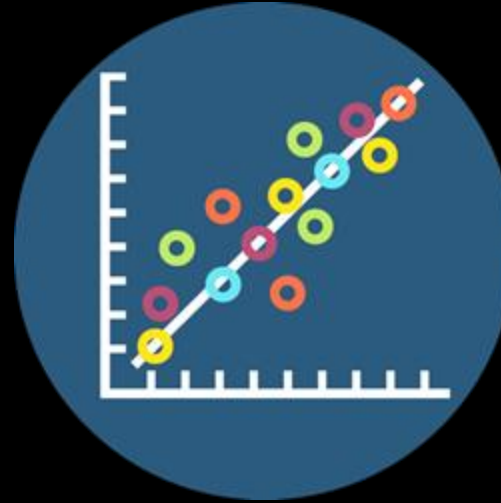
Research Methods



Observational
methods



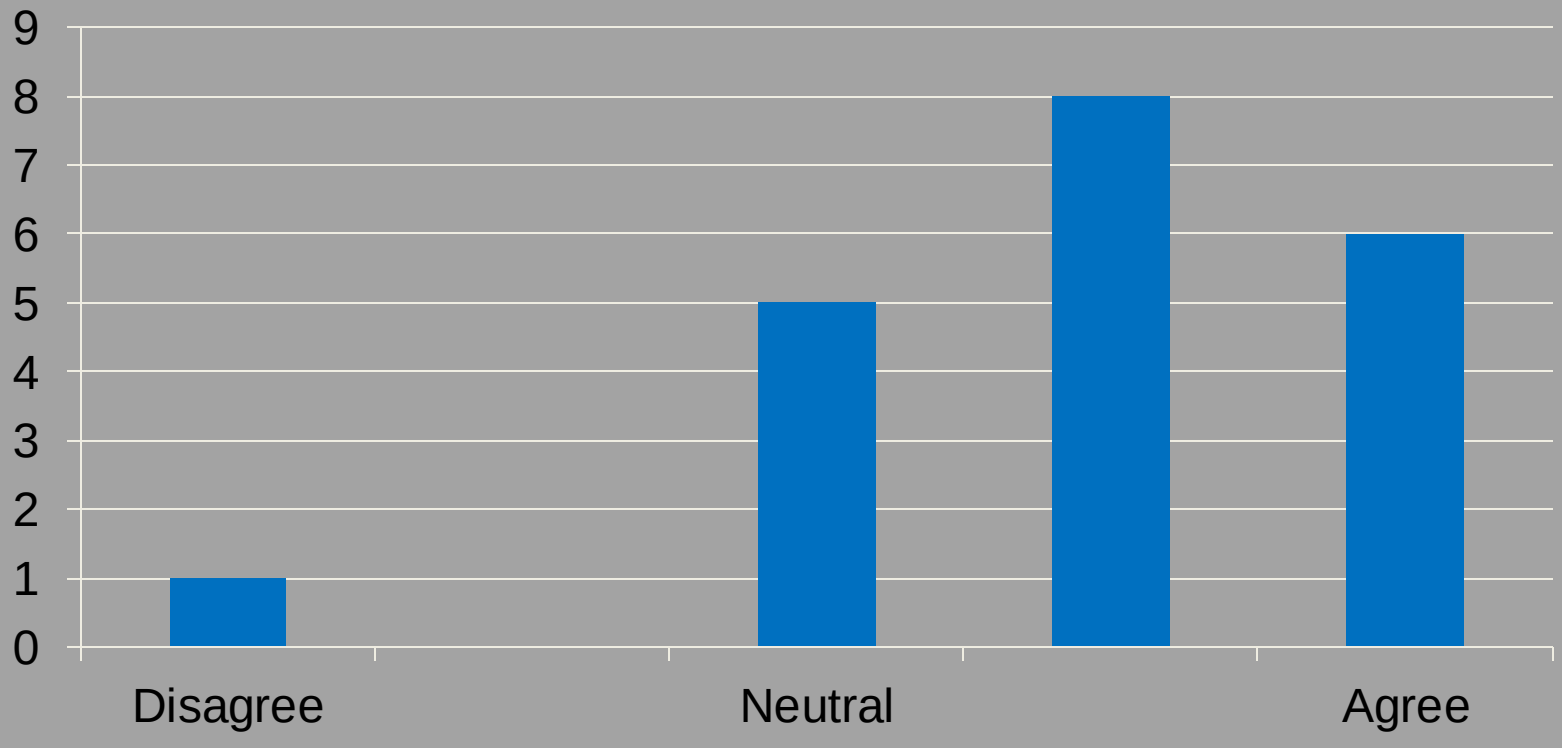
Experimental
methods



Correlational
methods

Why do we need controlled experiments?

Keyboard 1 is easy to use



Isolating the Cause and Effect

Participants rated the system easy to use, because

- they actually found the system easy to use
- they want to support you in your research
- they were overwhelmed by the system's novelty
- the Dutch football team won the world cup yesterday
- ...

Mere observation will **NOT help to find the answer!**

Controlled experiments are more reliable because

- They can **isolate cause and effect!**
- Knowing cause and effect allows **informed design**

Comparative Evaluation

Is Method B better than Method A?

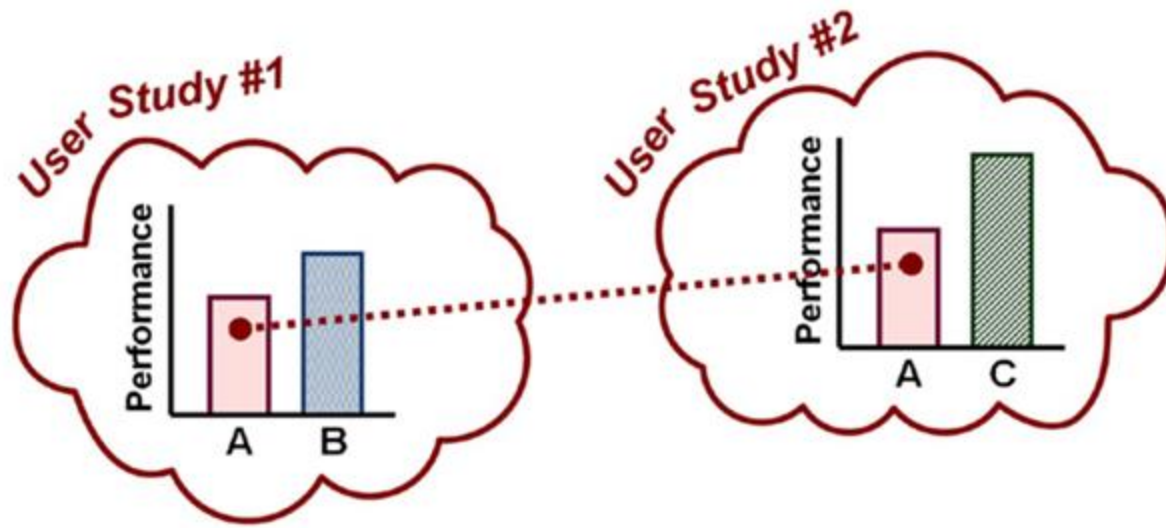
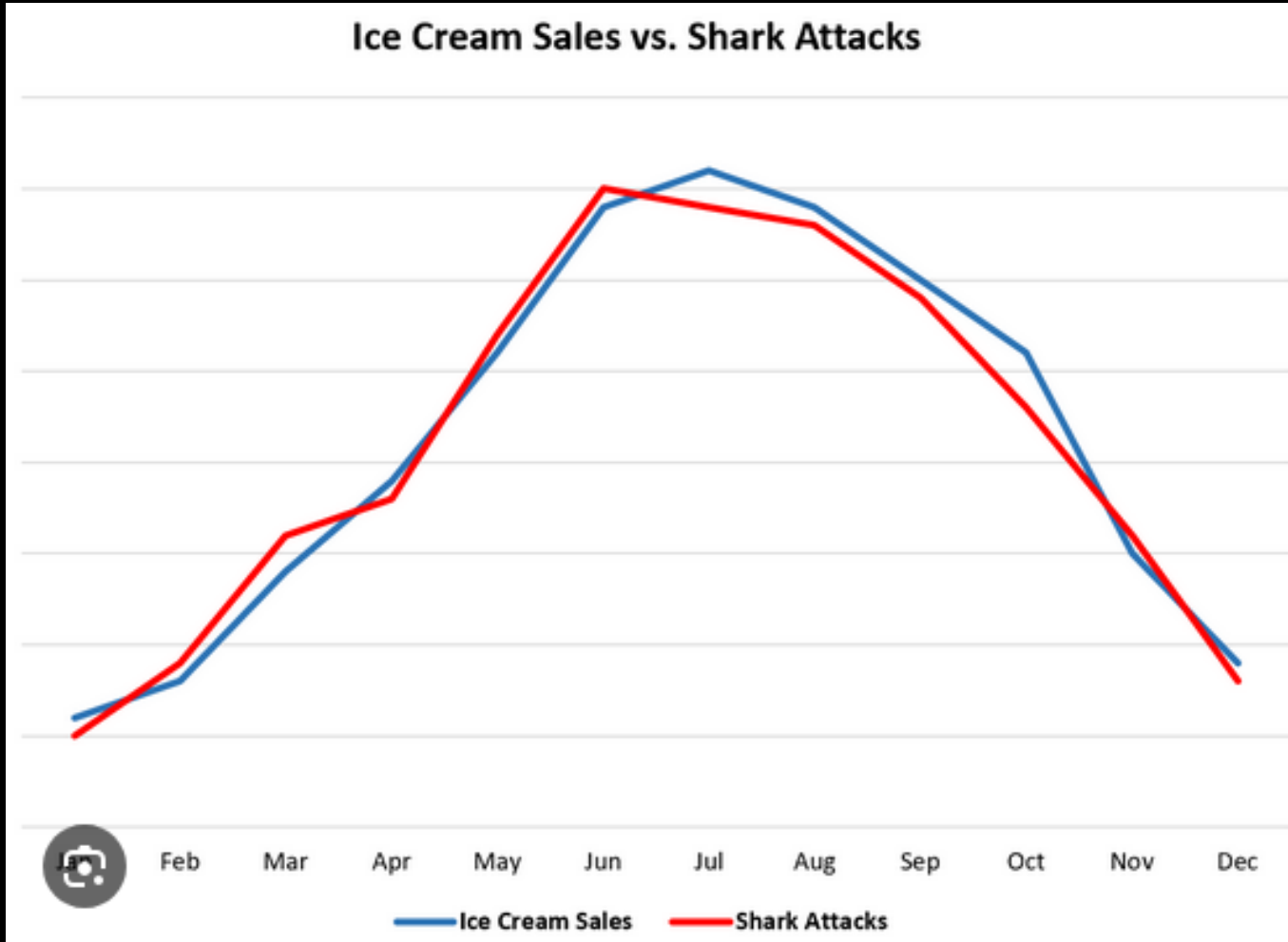


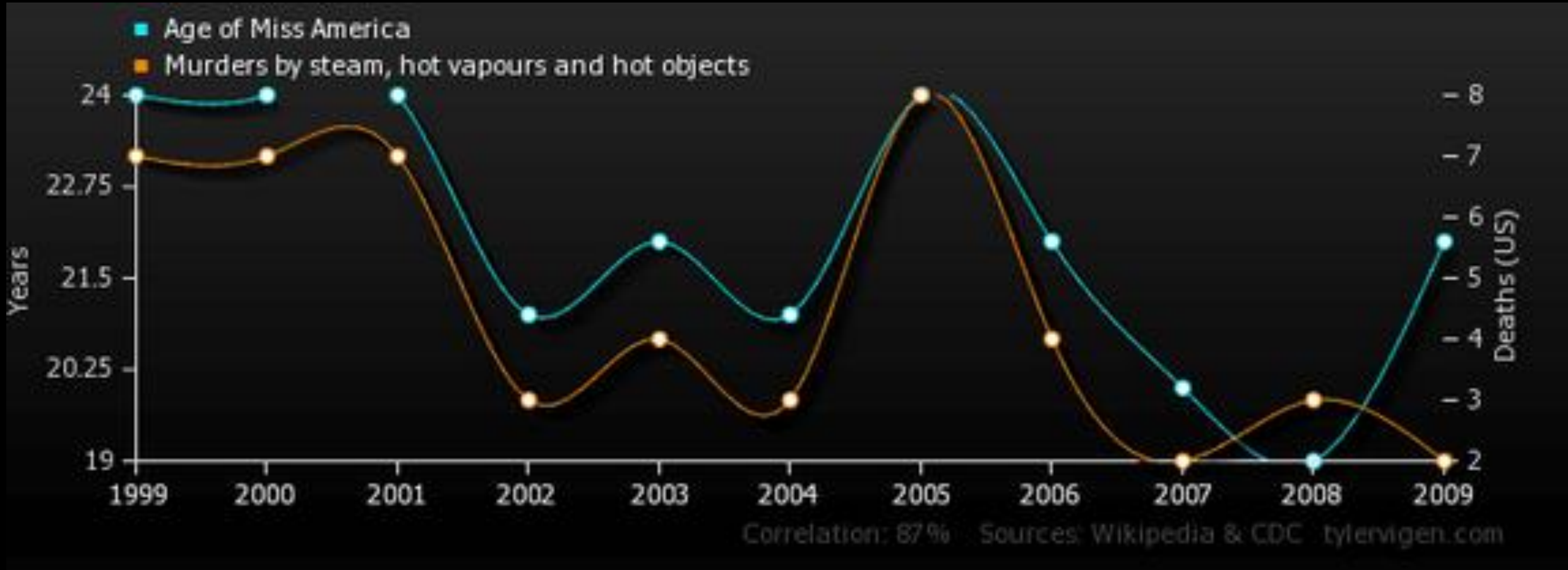
FIGURE 4.10

Including a baseline condition serves as a check on the methodology and facilitates the comparison of results between user studies.

Cause & Effect: Example 1

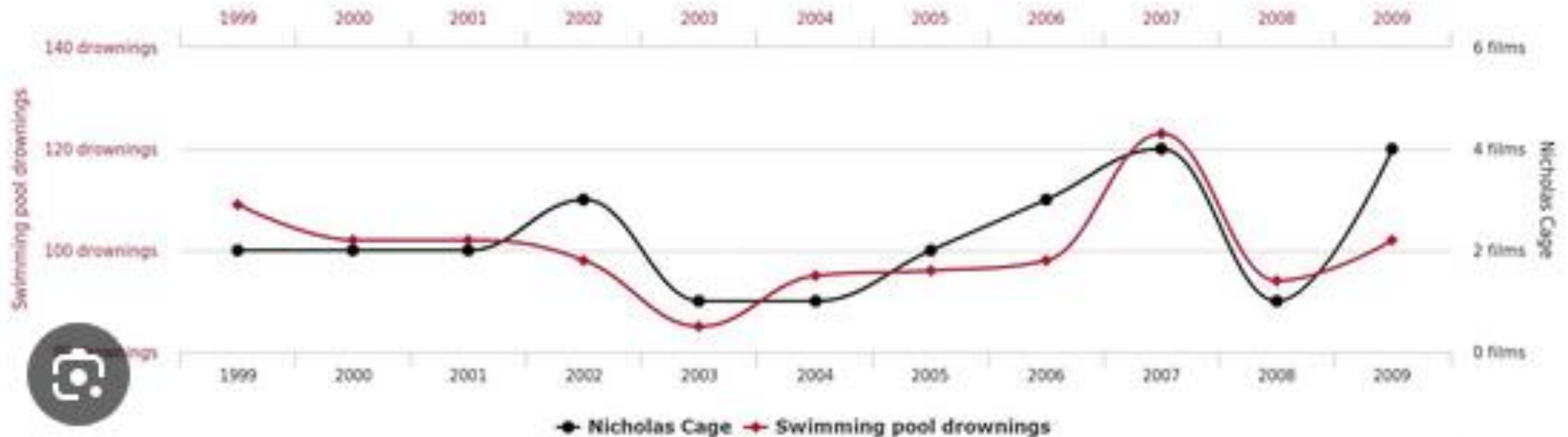


Cause & Effect: Example 2



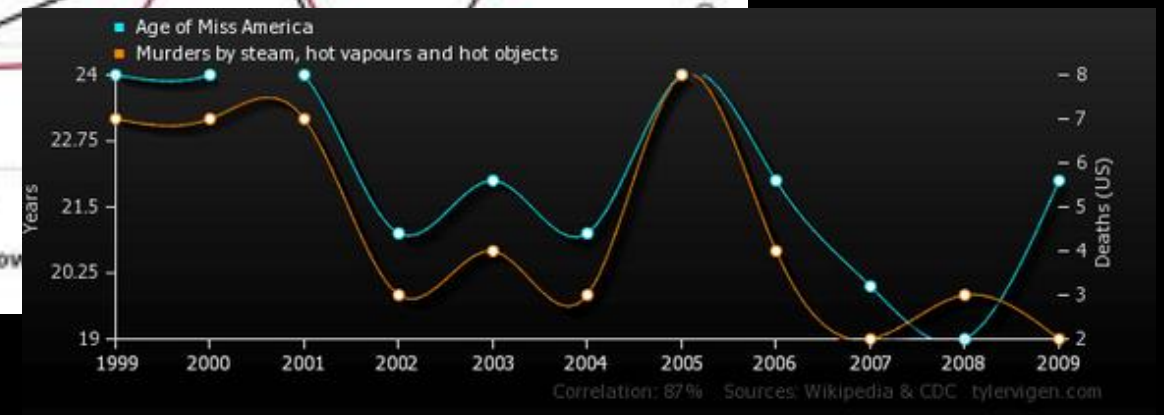
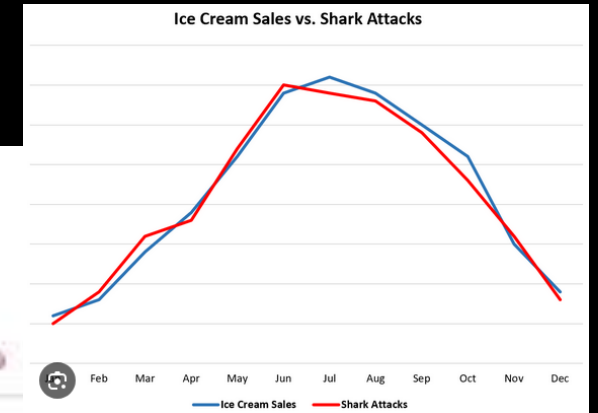
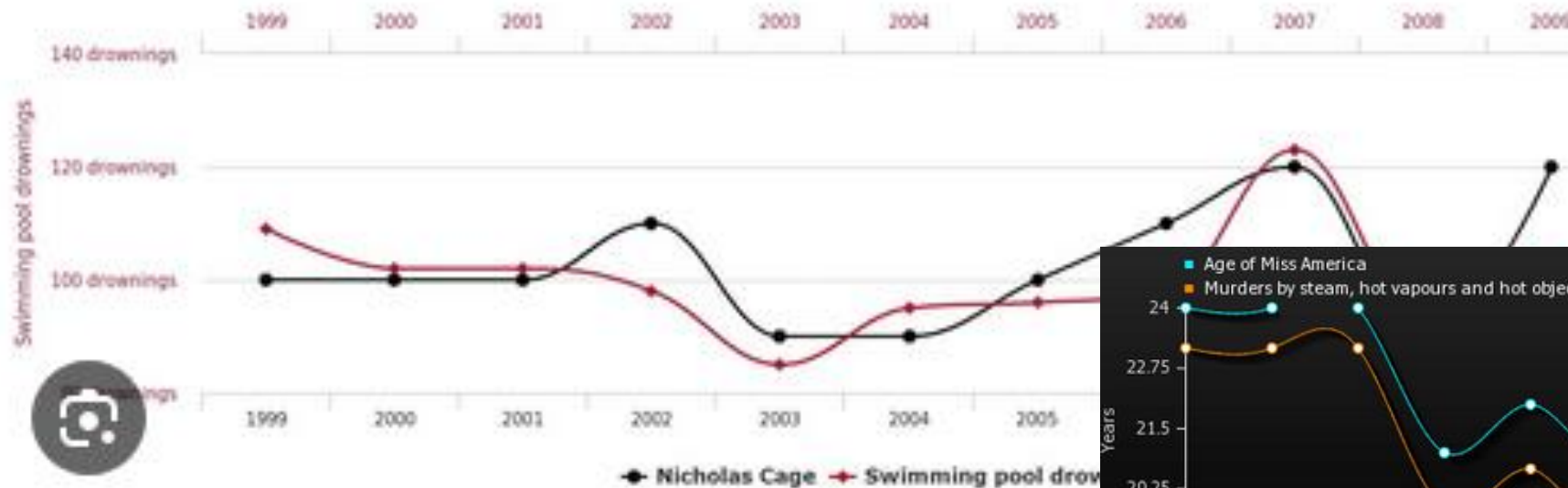
Cause & Effect: Example 3

Number of people who drowned by falling into a pool
correlates with
Films Nicolas Cage appeared in



What do these all have in common?

Number of people who drowned by falling into a pool
correlates with
Films Nicolas Cage appeared in



Correlation vs. Causation!

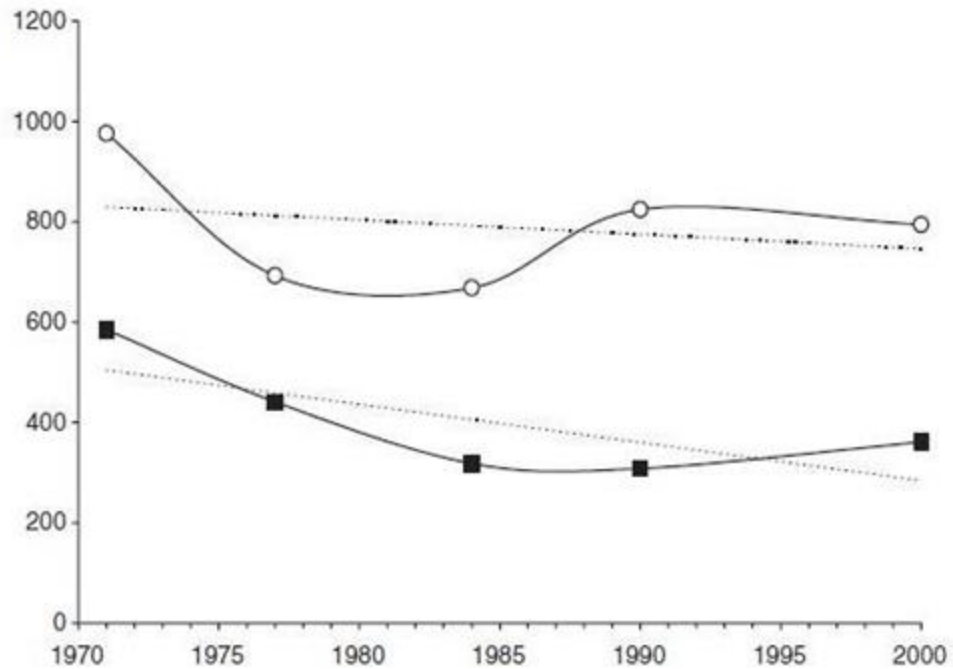


Figure 1. Storks and the birth rate in Lower Saxony, Germany (1971–2000). Open circles show yearly birthrates in hundreds in Lower Saxony. Full squares show numbers pairs of storks in Lower Saxony. Dotted lines represent linear regression trend ($y = mx + b$).

Correlation vs. Causation!

Storks Deliver Babies ($p = 0.008$)

KEYWORDS:

Teaching;
Correlation;
Significance;
p-values.

Robert Matthews

Aston University, Birmingham, England.
e-mail: rajm@compuserve.com

Summary

This article shows that a highly statistically significant correlation exists between stork populations and human birth rates across Europe. While storks may not deliver babies, unthinking interpretation of correlation and p -values can certainly deliver unreliable conclusions.

◆ INTRODUCTION ◆

Introductory statistics textbooks routinely warn of the dangers of confusing correlation with causation, pointing out that while a high correlation coefficient is indicative of (linear) association, it cannot be taken as a measure of causation. Such

association between storks and the concept of women as bringers of life, and also in the bird's feeding habits, which were once regarded as a search for embryonic life in water (Cooper 1992). The legend lives on to this day, with neonate-bearing storks being a regular feature of greetings cards celebrating births.

Correlation vs. Causation!

Birth rate and number of storks **correlate**

Explanation 1: children cause storks

- For example, the crying of babies attract storks

Explanation 2: storks cause children

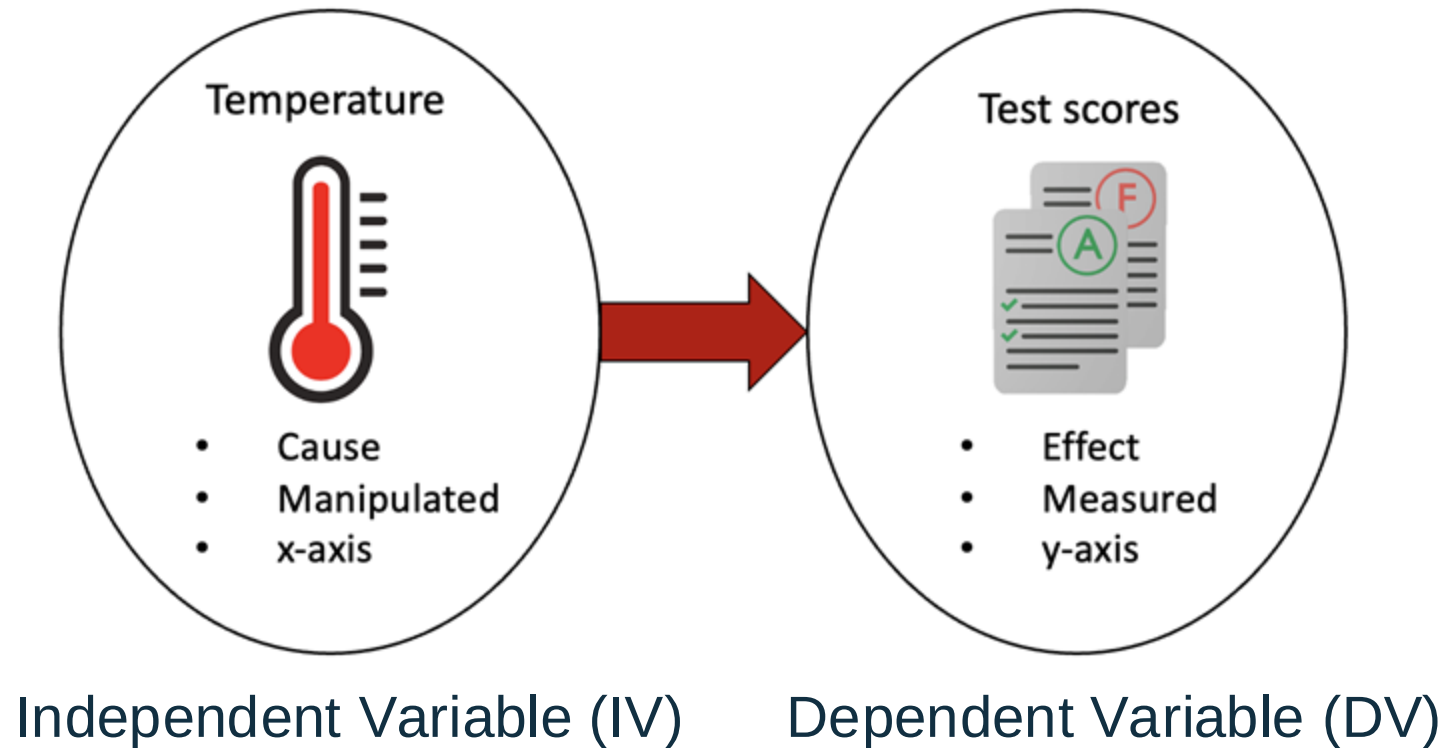
- For example, the myth is true and storks bring babies

Explanation 3: a third unknown aspect causes both

- For example, the village environment is more friendly to storks and families that desire children.
- **It's circumstantial**

Experiment in a Nutshell

Experiment: Does exam room temperature affect students test scores?

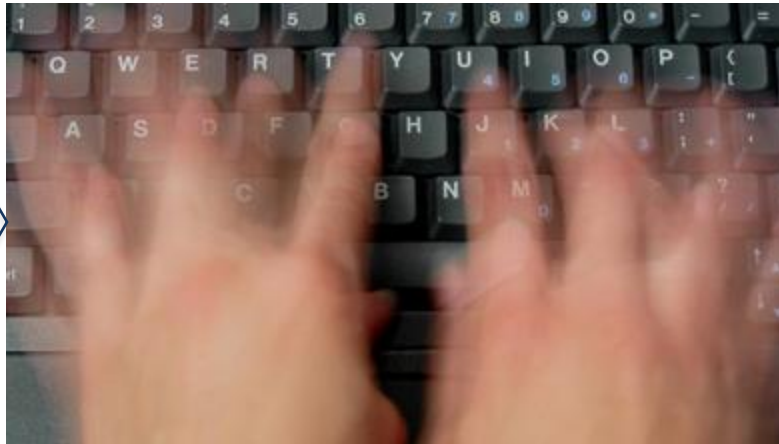


HCI experiments

Type of keyboard



Typing speed



Independent Variable (IV)

Dependent Variable (DV)

HCI experiments

Robot's facial expression



Learning gain



Independent Variable (IV)

Dependent Variable (DV)

HCI experiments



Technology



User



Context



Usability

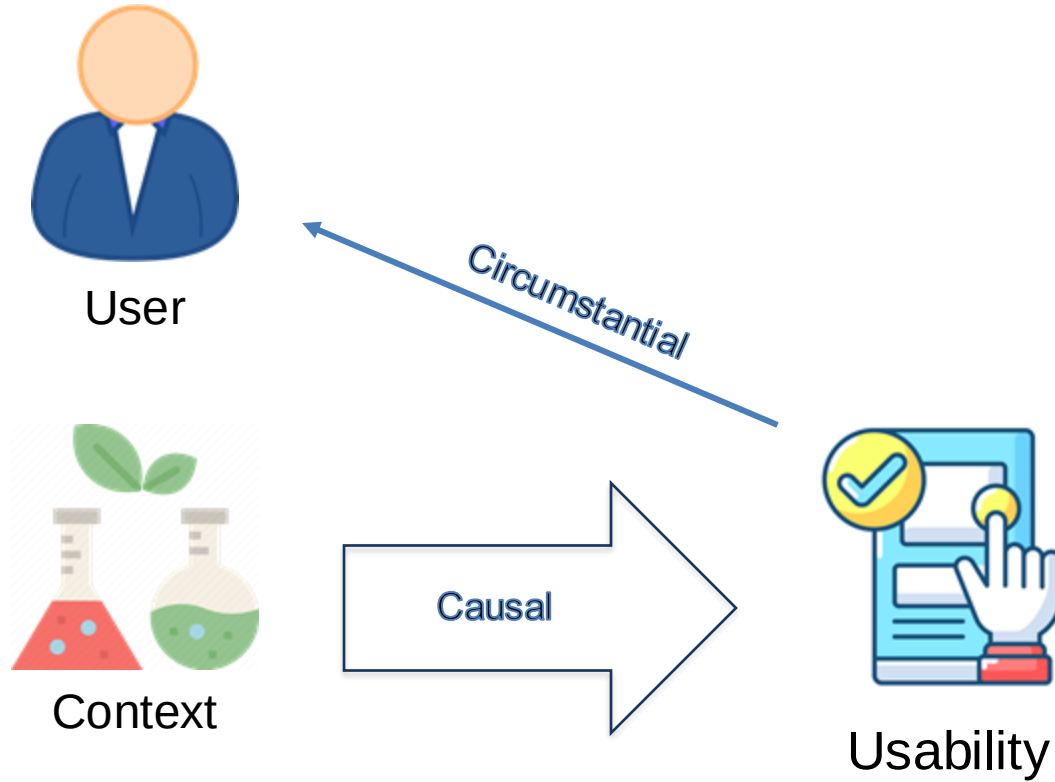


UX

Independent Variable (IV)

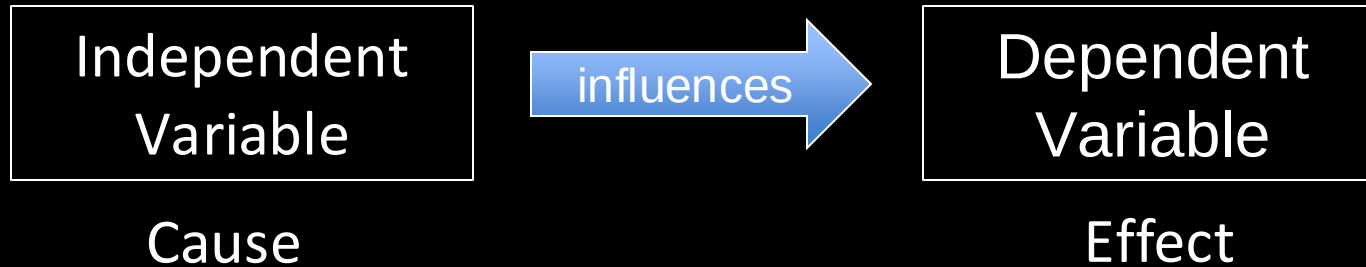
Dependent Variable (DV)

HCI experiments



Research Question (RQ) and Hypothesis

- A **Hypothesis** is a prediction about the relationship between two or more variables.
- A **RQ** is a specific question regarding the assumed relationship that the study tries to answer.



Research Question (RQ) and Hypothesis

- In HCI research, RQs are often about new or existing **user interface** or **interaction** technique.
- Variables can be any attribute or property of **humans**, **systems** or **environments**.

Example: Using keyboard X improves typing speed.

Hypothesis Operationalization

Hypotheses MUST be testable and measurable

- Keyboard X **is better than** keyboard Y.
Increases typing speed
- Adding element XX **improves usability** of the interface.
Reduces errors / increases user performance
- Adding auditory feedback in cars **improves situational awareness** in drivers.

Faster response to events in the driving simulator
(e.g., braking car / pedestrian crosses road)

How to formulate RQs/Hypotheses?

- **Literature review**

Know the state of the art

Identify the research gap

- **Think small**

Research must be practical and feasible

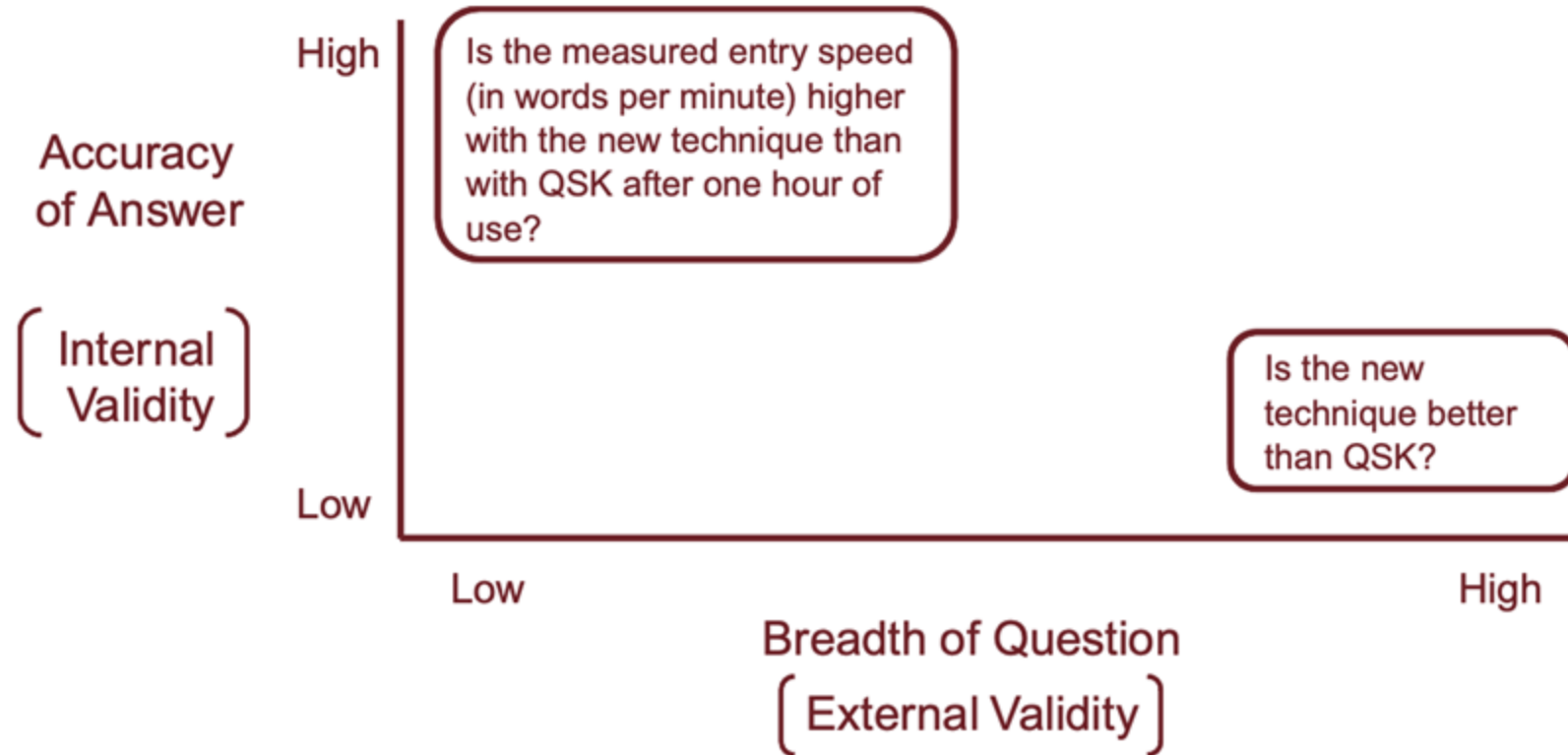
Instead of focusing on a *big idea*, find a small idea

- **Think inside the box!**

Get inspired by existing designs

Stick to established research methods

Internal vs. External Validity



Internal vs. External Validity

- degree of confidence in the causal relationship
- not influenced by other factors or variables
- generalizability of findings to other people, situations
- Requires large samples

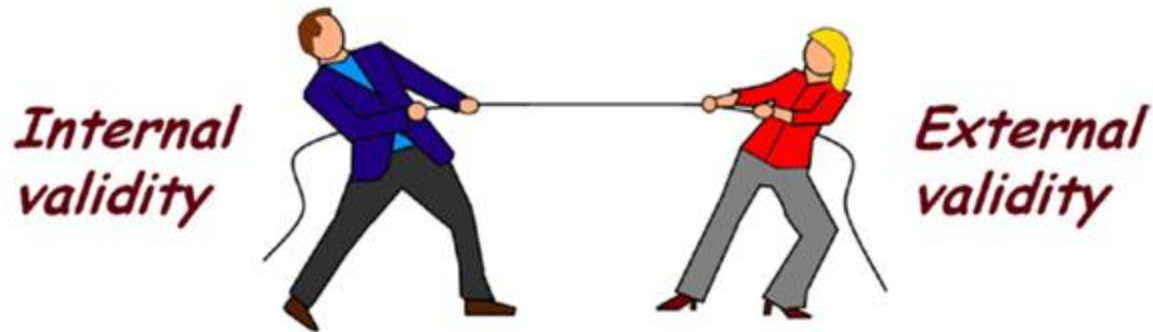
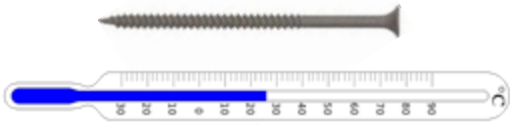


FIGURE 4.9

There is tension between internal validity and external validity. Improving one comes at the expense of the other.

Correct measurements

Length of
the screw?



intelligence?



<http://www.azune.com/hat-toque-sizes/>

Scales of measurement

Nominal:

- Mutually exclusive categories

Ordinal:

- Categories, but ordered by some attribute

Interval:

- Equal distance, no absolute zero

Ratio

- Equal distance and absolute zero

Example of measurements:

Car brands and measuring “Comfort”



Number of users of a specific car brand?

Nominal

Ranking best car to worst car wrt C&A?

Ordinal

“Seats are amazing!”:

Disagree (-1), Neutral (0) or Agree (1)?

Interval

How many seconds to turn on the car?

Ratio



Outcome eg: 100 Toyota cars vs 50 Ford

Problem eg: $\text{avg comfort} = 100 \text{ toyota} + 50 \text{ Ford} / 2$??

Outcome eg: $\text{avg rank of Toyota} > \text{avg rank of Ford}$

Problem eg:

How much better?

$\text{avg comfort} = ?$

Outcome eg:

avg value for Toyota: 0.8

avg value for Ford: 0.2

$\text{avg comfort} = 0.5$

Problem eg:

How much better is Toyota than Ford?

$0.8 - 0.2 = 0.5$? → None sense

Outcome eg:

Toyota, 1.2 sec on avg

Ford: 1.8 sec on avg

No problem! eg:

Avg time to turn on 1.5 sec

Toyota is ‘better’ than Ford by exactly ‘0.6 sec’

Mathematical properties of Scales of measurement

Nominal:

- No meaningful calculation

Ordinal:

- Comparison (less/greater-than), but no mean

Interval:

- Mean, but not ratios (eg. “20°C is twice as warm as 10°C”)

Ratio

- Universal: arithmetic operators (+, -, *, /), mean, std, var, ...
- Grounded in physical values (time, distance, velocity, number of actions, ...)

Scales of measurement

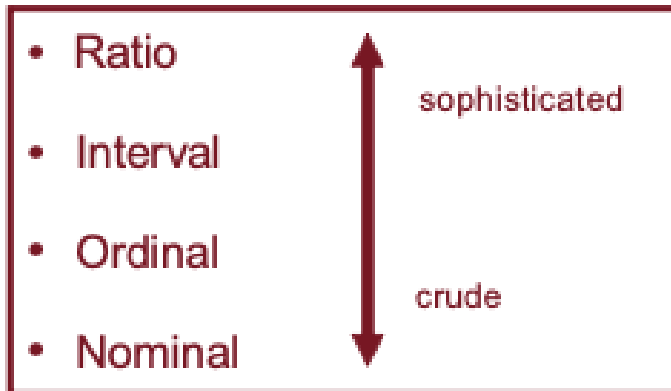


FIGURE 4.4

Scales of measurement: nominal, ordinal, interval, and ratio. Nominal measurements are considered simple, while ratio measurements are sophisticated.

Likert Scale

Please indicate your level of agreement with the following statements.

	Strongly disagree	Mildly disagree	Neutral	Mildly agree	Strongly agree
It is safe to talk on a mobile phone while driving.	1	2	3	4	5
It is safe to read a text message on a mobile phone while driving.	1	2	3	4	5
It is safe to compose a text message on a mobile phone while driving.	1	2	3	4	5

Likert Scale

Please indicate your level of agreement with the following statements.

	Strongly disagree	Mildly disagree	Neutral	Mildly agree	Strongly agree
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It is safe to compose a text message on a mobile phone while driving.	1	2	3	4	5

FIGURE 4.7

A set of questionnaire items organized in a Likert Scale. The responses are examples of interval scale data.

How to choose measurements?

- **Literature review**
 - Consensus in a scientific community
 - Validated scales and questionnaires
 - Results are comparable
- **If does not exist, define it for your experiment**
 - “For this experiment, we define driving experience as years of having a driver’s license.”
 - Less valid but quantified
 - Harder to compare to other results

Evaluation in HCI research

Consensus in HCI community

- Efficiency – Task completion time
- Accuracy – error rate
- Learnability – repeated measure of reaction time or performance
- ...

Standardized questionnaires

- Nasa-TLX (task load)
- AttrakDiff (attractiveness of UI)
- SUS (Usability)
- UTAUT (acceptance and use of UI)

NASA Task Load Index

Hart and Staveland's NASA Task Load Index (TLX) method assesses work load on five 7-point scales. Increments of high, medium and low estimates for each point result in 21 gradations on the scales.

Name	Task	Date

Mental Demand How mentally demanding was the task?

Very Low Very High

Physical Demand How physically demanding was the task?

Very Low Very High

Temporal Demand How hurried or rushed was the pace of the task?

Very Low Very High

Performance How successful were you in accomplishing what you were asked to do?

Perfect Failure

Effort How hard did you have to work to accomplish your level of performance?

Very Low Very High

Frustration How insecure, discouraged, irritated, stressed, and annoyed were you?

Very Low Very High

The Case of the iPhone



Academic Integrity: The Case of Prof Francesca Gino

Harvard Business School Prof. Gino
Accused of Plagiarism Following Data
Fraud Allegations



Break Time (15min)

Questions ?